

FORTIGATE NSE4

## Lab 4 — Routing Concepts

# 1. Overview

This lab builds a routed network in which a FortiGate and a Cisco router exchange routes using OSPF. It is deliberately multi-vendor — interoperability exposes the details that a single-vendor lab lets you skip.

## ► 1.1 Objectives

- Configure FortiGate interfaces with aliases, roles and administrative access
- Address the endpoints and verify Layer 3 connectivity to each gateway
- Configure OSPF area 0 on the FortiGate and on the Cisco router
- Set LAN interfaces passive so hellos are not sent onto user segments
- Tune the transit interface — network type and DR priority
- Verify the adjacency reaches FULL and routes are exchanged both ways
- Create firewall policies permitting inter-segment traffic
- Diagnose an end-to-end failure using ICMP codes, sniffer, and debug flow

## ► 1.2 Scope

**Dynamic routing is not part of the NSE 4 exam blueprint.** The Routing domain covers static routing and SD-WAN only. **This lab is built for practical skill and portfolio value rather than exam coverage.** What it does reinforce for the exam is the routing table, route selection, firewall policy behaviour, and the diagnostic workflow in section 9.

## ► 1.3 Routing Design

| Decision           | Value                            | Reasoning                                                                                        |
|--------------------|----------------------------------|--------------------------------------------------------------------------------------------------|
| Protocol           | OSPF, single area 0.0.0.0        | The backbone area — no area design complexity in a three-segment lab                             |
| Router IDs         | FortiGate 1.1.1.1, Cisco 2.2.2.2 | Set explicitly so the election outcome is readable rather than derived from an interface address |
| Transit type       | Broadcast                        | Both devices participate in the DR/BDR election                                                  |
| Passive interfaces | All LAN-facing ports             | See section 6.2                                                                                  |
| Redistribution     | None                             | Every network is advertised directly                                                             |

## 2. Topology and Addressing

A FortiGate and a Cisco vIOS router share a transit segment. The FortiGate owns two LAN segments, the router owns one. A Windows host on the transit network is used for management.

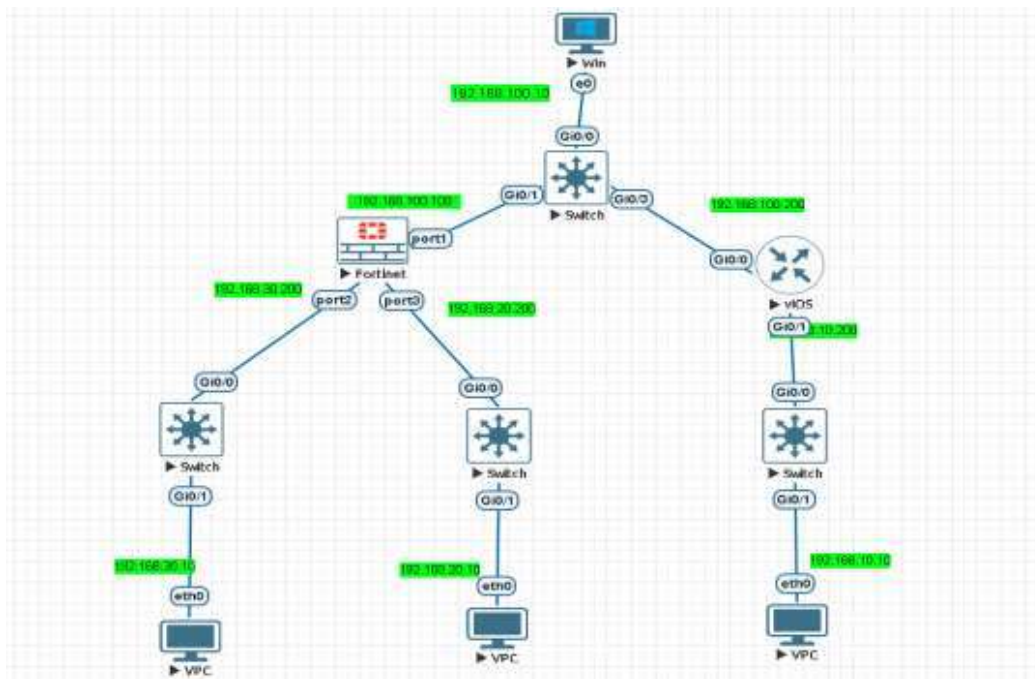


Figure 1 — EVE-NG lab topology

| Device     | Interface | Alias     | Address            | Segment        |
|------------|-----------|-----------|--------------------|----------------|
| FortiGate  | port1     | Transit   | 192.168.100.100/24 | Transit / mgmt |
| FortiGate  | port2     | 30-Subnet | 192.168.30.200/24  | LAN-30         |
| FortiGate  | port3     | 20 port   | 192.168.20.200/24  | LAN-20         |
| Cisco vIOS | Gi0/0     | —         | 192.168.100.200/24 | Transit        |
| Cisco vIOS | Gi0/1     | —         | 192.168.10.200/24  | LAN-10         |
| Win-MGMT   | e0        | —         | 192.168.100.10/24  | Management     |
| VPC-30     | eth0      | —         | 192.168.30.10/24   | GW .30.200     |
| VPC-20     | eth0      | —         | 192.168.20.10/24   | GW .20.200     |
| VPC-10     | eth0      | —         | 192.168.10.10/24   | GW .10.200     |

### 3. FortiGate Interfaces

Each interface is given an address, an alias, and the minimum administrative access it needs.

**Aliases are not cosmetic here. They appear throughout the OSPF interface table and the policy list**, so naming a port for its role rather than its number makes every later screen readable without cross-referencing.

```
edit "port1"
    set vdom "root"
    set ip 192.168.100.100 255.255.255.0
    set allowaccess ping https ssh http
    set type physical
    set snmp-index 1
```

Figure 2 — port1 — the transit interface

|                                                                                          |                                                                                                      |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Name                                                                                     |  port3              |
| Alias                                                                                    | 20 port                                                                                              |
| Type                                                                                     |  Physical Interface |
| VRF ID  | 0                                                                                                    |
| Role  | Undefined         |

☐ Dedicated Management Port

Address

|                      |                              |      |                      |                 |
|----------------------|------------------------------|------|----------------------|-----------------|
| Addressing mode      | Manual                       | DHCP | Auto-managed by IPAM | One-Arm Sniffer |
| IP/Netmask           | 192.168.20.200/255.255.255.0 |      |                      |                 |
| Secondary IP address | <input type="checkbox"/>     |      |                      |                 |

Administrative Access

|      |                                            |                                                                                                                                         |                                     |
|------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| IPv4 | <input type="checkbox"/> HTTPS             | <input checked="" type="checkbox"/> PING                                                                                                | <input type="checkbox"/> FMG-Access |
|      | <input type="checkbox"/> SSH               | <input type="checkbox"/> SNMP                                                                                                           | <input type="checkbox"/> FTM        |
|      | <input type="checkbox"/> RADIUS Accounting | <input type="checkbox"/> Security Fabric Connection  | <input type="checkbox"/> Speed Test |

Figure 3 — port3 configured as "20 port"

Name: 30-Subnet (port2)

Alias: 30-Subnet

Type: Physical Interface

VRF ID: 0

Role: LAN

☐ Dedicated Management Port

Address

Addressing mode: Manual | DHCP | Auto-managed by IPAM | One-Arm Sniffer

IP/Netmask: 192.168.30.200/255.255.255.0

Create address object matching subnet: ☐

Secondary IP address: ☐

Administrative Access

IPv4

☐ HTTPS ☒ PING ☐ FMG-Access

☐ SSH ☐ SNMP ☐ FTM

☐ RADIUS Accounting ☐ Security Fabric Connection ☐ Speed Test

Figure 4 — port2 configured as "30-Subnet" with role LAN

Administrative access is scoped per interface: HTTPS and SSH on the transit segment where management lives, PING only on the LAN-facing ports.

## 4. Endpoint Configuration

Each VPCS node is addressed and given a default gateway, then verified against that gateway before any routing is attempted.

```
# VPC-30
ip 192.168.30.10/24 192.168.30.200
# VPC-20
ip 192.168.20.10/24 192.168.20.200
# VPC-10
ip 192.168.10.10/24 192.168.10.200
save
```

```
VPCS> ip 192.168.30.10/24 192.168.30.200
Checking for duplicate address...
VPCS : 192.168.30.10 255.255.255.0 gateway 192.168.30.200
```

Figure 5 — VPC-30 addressed with its gateway

```
VPCS> ping 192.168.30.200

84 bytes from 192.168.30.200 icmp_seq=1 ttl=255 time=1.594 ms
84 bytes from 192.168.30.200 icmp_seq=2 ttl=255 time=1.471 ms
84 bytes from 192.168.30.200 icmp_seq=3 ttl=255 time=1.382 ms
84 bytes from 192.168.30.200 icmp_seq=4 ttl=255 time=1.730 ms
84 bytes from 192.168.30.200 icmp_seq=5 ttl=255 time=1.507 ms
```

Figure 6 — VPC-30 reaches its gateway — the local segment is healthy

```
VPCS> ip 192.168.10.10/24 192.168.10.200
Checking for duplicate address...
VPCS : 192.168.10.10 255.255.255.0 gateway 192.168.10.200
```

Figure 7 — VPC-10 addressed with its gateway

**Testing bottom-up means later failures cannot be blamed on the endpoints.** Each host proves it can reach its own gateway before the routing protocol is introduced, so a subsequent failure belongs to routing or policy by elimination.

#### ► 4.1 VPCS Addressing Is Not Persistent

VPCS loses its addressing on restart unless `save` is run. **A node that silently reverted to no gateway will accept pings and never reply to an off-subnet source** — producing a timeout that looks exactly like a firewall problem. This caused one of the failures in section 9.

## 5. Cisco Router Configuration

```
interface GigabitEthernet0/0
 ip address 192.168.100.200 255.255.255.0
 no shutdown
!
interface GigabitEthernet0/1
 ip address 192.168.10.200 255.255.255.0
 no shutdown
!
router ospf 1
 router-id 2.2.2.2
 network 192.168.100.0 0.0.0.255 area 0
 network 192.168.10.0 0.0.0.255 area 0
 passive-interface GigabitEthernet0/1
```

```
Router#show interfaces summary
*Aug  8 20:12:40.400: %SYS-5-CONFIG_I: Configured from p interface brief
Interface                IP-Address      OK? Method Status
GigabitEthernet0/0       192.168.100.200 YES manual up
GigabitEthernet0/1       192.168.10.200  YES manual up
GigabitEthernet0/2       unassigned      YES unset  administratively down
```

Figure 8 — Both interfaces up/up

```
Router(config)#router ospf 1
Router(config-router)#router-id 2.2.2.2
Router(config-router)#network 192.168.100.0 0.0.0.255 area 0
Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
```

Figure 9 — OSPF process 1, router-id 2.2.2.2, both networks in area 0

## ► 5.1 Passive Does Not Mean Unadvertised

This behaves differently to what the name suggests. `passive-interface` stops OSPF hellos on that interface, but **the network is still advertised into OSPF provided a matching network statement exists**. Passive suppresses adjacency formation, not advertisement. The same is true on FortiGate.

## ► 5.2 Wildcard Masks, Not Subnet Masks

Cisco network statements take a wildcard mask — `0.0.0.255` for a /24, not `255.255.255.0`.

**Getting this backwards is one of the most common reasons a network silently fails to be advertised**, and the router accepts the statement without complaint.

# 6. FortiGate OSPF

Router ID

Areas

| Area ID | Type    | Authentication |
|---------|---------|----------------|
| 0.0.0.0 | Regular | None           |

1

Networks

| Network          | Area    |
|------------------|---------|
| 192.168.100.0/24 | 0.0.0.0 |
| 192.168.20.0/24  | 0.0.0.0 |
| 192.168.30.0/24  | 0.0.0.0 |

3

Figure 10 — Router ID 1.1.1.1, area 0.0.0.0, three networks advertised

| Interfaces                                                               |                   |      |             |                |         |  |
|--------------------------------------------------------------------------|-------------------|------|-------------|----------------|---------|--|
| <div> <div>+ Create New</div> <div> Edit</div> <div> Delete</div> </div> |                   |      |             |                |         |  |
| Name                                                                     | Interfaces        | Cost | Apply To IP | Authentication | Passive |  |
| port2                                                                    | 30-Subnet (port2) | 0    | Any IP      | None           | Enabled |  |
| port3                                                                    | 20-port (port3)   | 0    | Any IP      | None           | Enabled |  |

Figure 11 — port2 and port3 set passive

## ► 6.1 Transit Interface Tuning

The transit segment is a broadcast network, so a DR/BDR election takes place. Setting the network type and priority explicitly makes the outcome deterministic rather than incidental.

```
config router ospf
  config ospf-interface
    edit "Transit"
      set interface "port1"
      set network-type broadcast
      set priority 10
    next
  end
end
```

```
Fortigate-1 # config router ospf
Fortigate-1 (ospf) # config ospf-interface
Fortigate-1 (ospf-interface) # edit "Transit"
new entry 'Transit' added
Fortigate-1 (Transit) # set interface "port1"
Fortigate-1 (Transit) # set network-type broadcast
Fortigate-1 (Transit) # set priority 10
Fortigate-1 (Transit) # next
Fortigate-1 (ospf-interface) # end
Fortigate-1 (ospf) # end
```

Figure 12 — Transit interface — broadcast network type, priority 10

## ► 6.2 Why the LANs Must Be Passive

A network statement enables OSPF on *every* interface whose subnet matches — including LAN-20 and LAN-30, where only endpoints live.

**Without passive-interface, the firewall sends hellos onto user segments and will accept an adjacency from anything that answers.** That is unnecessary attack surface, and in an OT or segmented environment it is an audit finding.

## 7. Firewall Policies

**A route tells the FortiGate where to send a packet. A policy decides whether it may.** Both are required, and **OSPF forming an adjacency proves nothing about user traffic** — the control plane and the data plane are permitted separately.



Figure 13 — Policies permitting traffic between the LAN and the transit interface

### ► 7.1 NAT Must Be Disabled on Internal Policies

NAT is enabled by default on new policies. Between internal segments that is wrong: it rewrites 192.168.30.10 to the firewall's transit address, so the peer never sees the real source and every log entry loses attribution.

**Worse, it can mask a missing return route. Traffic appears to work because the translated source is directly connected to the peer** — so the routing fault stays hidden until NAT is removed or a different source is tested.

### ► 7.2 Both Directions Are Required

An ICMP echo needs a policy outbound and the echo-reply needs one inbound. A single policy produces a silent timeout that is easily mistaken for a routing failure.

*port3 (LAN-20) has no policies in the configuration above, so VPC-20 remains isolated by design. Repeat the pair using port3, or place port2 and port3 in a zone and write one policy pair against it — the cleaner approach as segments grow.*

## 8. Verification

**Verify in layers: adjacency, then route exchange on both sides, then end-to-end traffic.** A failure at any one layer localises the fault immediately, rather than leaving the whole path under suspicion.

## ► 8.1 Transit Reachability

```
execute ping 192.168.100.200
```

```
Fortigate-1 # execute ping 192.168.100.200
PING 192.168.100.200 (192.168.100.200): 56 data bytes
64 bytes from 192.168.100.200: icmp_seq=0 ttl=255 time=6.7 ms
64 bytes from 192.168.100.200: icmp_seq=1 ttl=255 time=2.1 ms
64 bytes from 192.168.100.200: icmp_seq=2 ttl=255 time=1.9 ms
64 bytes from 192.168.100.200: icmp_seq=3 ttl=255 time=2.0 ms
64 bytes from 192.168.100.200: icmp_seq=4 ttl=255 time=2.2 ms
```

Figure 14 — The FortiGate reaches the Cisco router across the transit segment

## ► 8.2 OSPF Adjacency

```
get router info ospf neighbor
```

State **FULL** confirms a great deal at once: **hellos in both directions, matching area, matching hello and dead timers, matching MTU, and matching authentication**. A single status word rules out five separate misconfigurations.

## ► 8.3 Routes Learned on the FortiGate

```
get router info routing-table all
```

```
Fortigate-1 # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4
       * - candidate default

Routing table for VRF=0
O       192.168.10.0/24 [110/2] via 192.168.100.200, port1, 00:06:11, [1/0]
C       192.168.20.0/24 is directly connected, port3
C       192.168.30.0/24 is directly connected, port2
C       192.168.100.0/24 is directly connected, port1
```

Figure 15 — O 192.168.10.0/24 [110/2] via 192.168.100.200, port1

The **o** code marks an OSPF route and **110** is the OSPF administrative distance. The **c** entries are the directly connected networks.

## ► 8.4 No Competing Static Routes

```
show router static
```

```
Fortigate-1 # show router static
config router static
end
```

Figure 16 — Static route table empty — nothing competes with OSPF

**This check matters more than it looks. Static routes have an administrative distance of 10 against OSPF's 110, so any leftover static route to the same prefix silently wins and OSPF never installs.** Disabling redistribution does not remove the route itself — it must be deleted.

## ► 8.5 Routes Learned on the Cisco Router

```
show ip route ospf
```

```
Router>show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O        192.168.20.0/24
         [110/2] via 192.168.100.100, 00:20:54, GigabitEthernet0/0
O        192.168.30.0/24
         [110/2] via 192.168.100.100, 00:20:54, GigabitEthernet0/0
```

Figure 17 — The router has learned both FortiGate LANs via 192.168.100.100

Route exchange must be confirmed in both directions. A one-way exchange produces a request that arrives and a reply that never returns.

## ► 8.6 End-to-End Test

```
# from VPC-30
ping 192.168.10.10
```

```
VPCS> ping 192.168.10.10
84 bytes from 192.168.10.10 icmp_seq=1 ttl=62 time=12.563 ms
84 bytes from 192.168.10.10 icmp_seq=2 ttl=62 time=11.371 ms
84 bytes from 192.168.10.10 icmp_seq=3 ttl=62 time=7.334 ms
84 bytes from 192.168.10.10 icmp_seq=4 ttl=62 time=11.802 ms
84 bytes from 192.168.10.10 icmp_seq=5 ttl=62 time=16.027 ms
```

Figure 18 — End-to-end success — replies with TTL 62

**Reading the TTL.** VPCS sends with TTL 64. **A reply at 62 means two routing hops — the FortiGate and the Cisco router — which is exactly the expected path.** TTL is a free sanity check that traffic took the route you think it did.

## 9. Troubleshooting

The lab did not work first time. The sequence of failures below is the most valuable part of the exercise — each had a distinct signature that narrowed the fault before any configuration was changed.

### ► 9.1 Failure 1 — Network Unreachable

The first attempts returned ICMP type 3, code 0 — network unreachable — generated by the FortiGate itself. **That message only appears when the device has no route to the destination, which immediately eliminated firewall policy as a cause.**

| Ping symptom                     | Layer at fault     | Where to look                               |
|----------------------------------|--------------------|---------------------------------------------|
| ICMP type 3 code 0 — unreachable | Routing            | Routing table, OSPF adjacency, LSA exchange |
| Silent timeout, no reply at all  | Policy or endpoint | Firewall policy, endpoint gateway           |
| Replies with unexpected TTL      | Path               | Asymmetric or longer route than expected    |

### ► 9.2 Failure 2 — Silent Timeout

Once OSPF installed the route, the error changed from unreachable to a plain timeout. **That transition is itself the diagnosis** — routing was fixed, and the fault had moved downstream to policy or the endpoints.

The cause was an endpoint: VPC-10 had no default gateway configured, so it received each echo request but could not route the reply to an off-subnet source.

### ► 9.3 Failure 3 — Confirming with Debug Flow and Sniffer

**The two tools answer different questions.** Debug flow shows the route lookup, the policy match and the drop reason; the sniffer shows whether the packet ever reached the wire. **Used together they distinguish an internal drop from an endpoint that is simply not replying.**

## 10. Diagnostic Reference

| Command                                           | Answers                                                           |
|---------------------------------------------------|-------------------------------------------------------------------|
| <code>get router info ospf neighbor</code>        | Is the adjacency FULL?                                            |
| <code>get router info ospf interface</code>       | Which interfaces run OSPF, which are passive, and the DR/BDR role |
| <code>get router info ospf database</code>        | Which LSAs were received — is the peer advertising the prefix?    |
| <code>get router info routing-table all</code>    | What is installed in the forwarding table                         |
| <code>show router static</code>                   | Is a static route beating OSPF?                                   |
| <code>diagnose sniffer packet &lt;intf&gt;</code> | Did the packet enter or leave that interface?                     |
| <code>diagnose debug flow</code>                  | Route lookup, policy match, drop reason                           |
| <code>diagnose sys session clear</code>           | Clears stale sessions created while routing was broken            |
| <code>show ip route ospf (Cisco)</code>           | Did the peer learn our networks?                                  |

## 11. Summary

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| Component          | Detail                                                                        |
|--------------------|-------------------------------------------------------------------------------|
| Protocol           | OSPF, single area 0.0.0.0                                                     |
| Router IDs         | FortiGate 1.1.1.1, Cisco 2.2.2.2                                              |
| Transit segment    | 192.168.100.0/24 — broadcast network type, FortiGate priority 10              |
| FortiGate LANs     | 192.168.20.0/24 and 192.168.30.0/24                                           |
| Cisco LAN          | 192.168.10.0/24                                                               |
| Passive interfaces | All LAN-facing ports on both devices                                          |
| Firewall policies  | Bidirectional pair between port2 and port1, NAT disabled, all sessions logged |
| Adjacency          | FULL — confirming hellos, area, timers, MTU and authentication all match      |
| Route exchange     | Confirmed in both directions                                                  |
| End-to-end result  | TTL 62 — two routing hops, the expected path                                  |
| Faults encountered | No route installed; endpoint missing its default gateway                      |